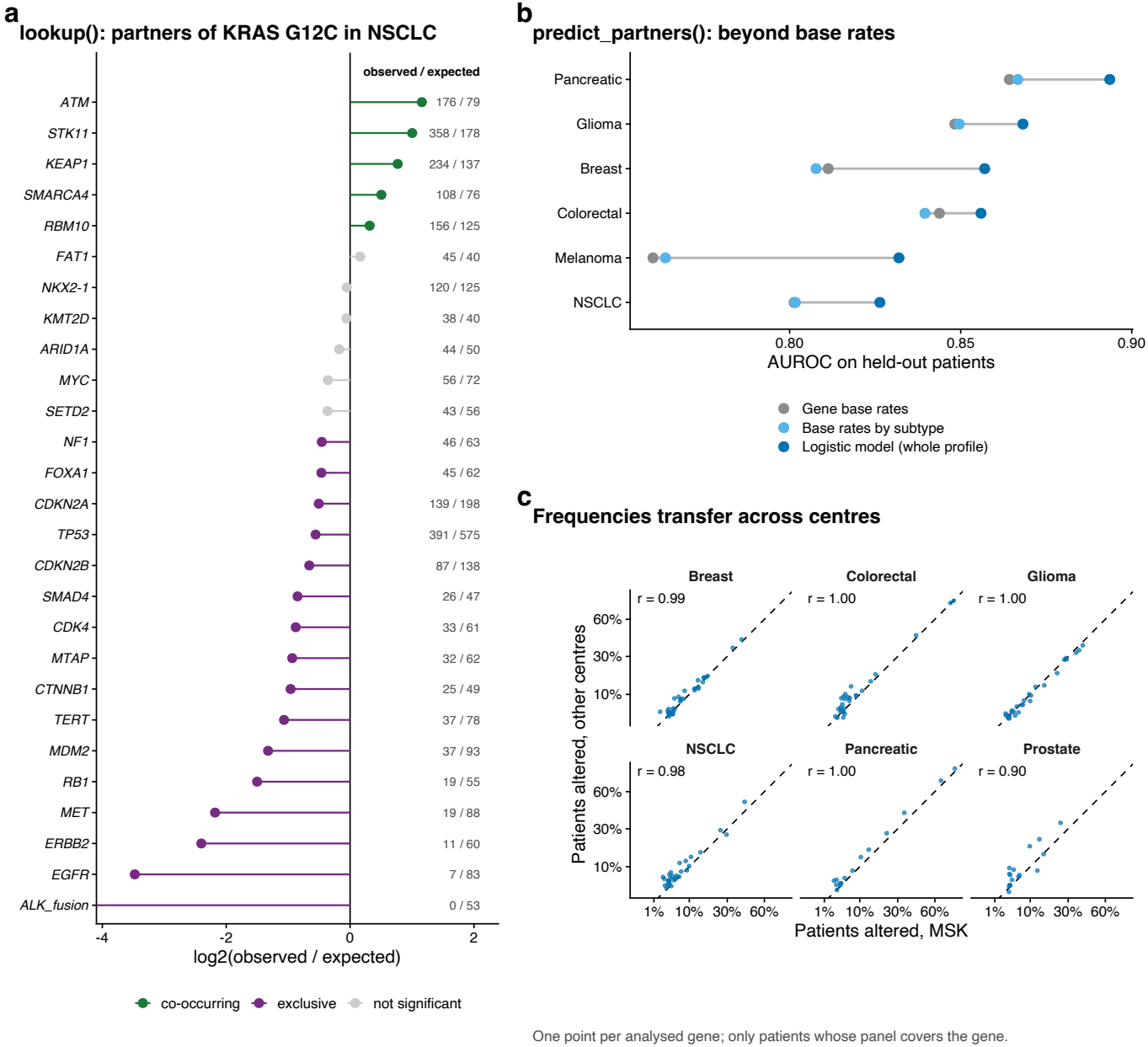
a
lookup(): partners of KRAS G12C in NSCLC
observed / expected
ATM 176 / 79
STK11 358 / 178
KEAP1 234 / 137
SMARCA4 108 / 76
RBM10 156 / 125
FAT1 45 / 40
NKX2-1 120 / 125
KMT2D 38 / 40
ARID1A 44 / 50
MYC 56 / 72
SETD2 43 / 56
NF1 46 / 63
FOXA1 45 / 62
CDKN2A 139 / 198
TP53 391 / 575
CDKN2B 87 / 138
SMAD4 26 / 47
CDK4 33 / 61
MTAP 32 / 62
CTNNB1 25 / 49
TERT 37 / 78
MDM2 37 / 93
RB1 19 / 55
MET 19 / 88
ERBB2 11 / 60
EGFR 7 / 83
ALK_fusion 0 / 53
-4
-2
0
2
log2(observed / expected)
co-occurring
exclusive
not significant
b
predict_partners(): beyond base rates
Pancreatic
Glioma
Breast
Colorectal
Melanoma
NSCLC
0.80
0.85
0.90
AUROC on held-out patients
Gene base rates
Base rates by subtype
Logistic model (whole profile)
c
Frequencies transfer across centres
Breast
Colorectal
Glioma
NSCLC
Pancreatic
Prostate
r = 0.99
r = 1.00
r = 1.00
r = 0.98
r = 1.00
r = 0.90
60%
30%
10%
1%
10%
30%
60%
Patients altered, other centres
Patients altered, MSK
One point per analysed gene; only patients whose panel covers the gene.